A three-layer framework for assessing disclosure-materiality misalignment

fixed structure + configurable components → reusable across industries, corpora, and language models

fixed (framework structure) · ⚙ configurable component · ✓ robustness-checked component

Layer 1 Corpus and benchmark definition

Select firm reports & reporting years

firms × years \u2192 documents $\{r_1, \ldots, r_m\}$

e.g. annual sustainability reports · integrated reports · 10-K / annual reports · CDP / TCFD filings

Select industry materiality framework

e.g. SASB · GRI · ESRS · TCFD · TNFD · CSRD

benchmark topics $F = \{f_1, f_2, \ldots, f_n\}$

f_1 | f_2 | f_3 | f_4 | … | f_n

Layer 2 Evidence retrieval, disclosure scoring, and MMRI aggregation

Extract topic-relevant evidence

per document — configurable: PDF text extractor · passage splitter · topic retriever

⚙ **tool examples**

pdfplumber · PyMuPDF · OCR · keyword · BM25 · embedding-based · hybrid retrieval

top-K passages per topic

Score topic-level disclosure coverage

per topic — configurable: language model, prompting approach, weights α, β, γ

⚙ **language model**

Claude · Gemini · OpenAI GPT · Llama

⚙ **prompting approach**

zero-shot · few-shot · chain-of-thought · structured JSON

⚙ **coverage weights α, β, γ (within a topic)**

equal $\alpha = \beta = \gamma = 1/3$ · domain-tuned · stakeholder-elicited

coverage score (fixed)

$$C_{r,i} = \alpha\,\mathrm{Pres} + \beta\,\mathrm{Spec} + \gamma\,\mathrm{Quant}$$
$$\alpha + \beta + \gamma = 1, \quad C_{r,i} \in [0,1]$$

✓ **robustness checks**
reference set · repeated k runs · ablation over α, β, γ / prompt / model

$C_{r,i}$

Convert to misalignment

per topic

misalignment (fixed)

$$M_{r,i} = 1 - C_{r,i} \in [0,1]$$
0 = fully disclosed
1 = silent on required topic

⚙ **applicability $a_i \in \{0,1\}$**

marks topics that do not apply to every firm

$M_{r,i}$

Aggregate to MMRI

across topics \u2014 configurable: applicability a_i, topic weights w_i

aggregation formula (fixed)

$$\mathrm{MMRI}_r = \frac{\sum_i a_i w_i M_{r,i}}{\sum_i a_i w_i} \in [0,1]$$
$$a_i \in \{0,1\}, \quad \sum_i w_i = 1$$

report mean $\bar{\mathrm{MMRI}}_r$ with 95% CI over k runs

⚙ **topic weights w_i (across topics)**

equal $w_i = 1/n$ · investor-survey · severity-adjusted

✓ **robustness checks**
repeated k runs · ablation over w_i · ensemble across language models

repeated for each materiality topic

comparable risk signal

Layer 3 Decision-relevant outputs

Per-document heatmap

$M_{r,i}$ across all documents
(firm × period × topic)

Topic-level profile

mean misalignment per firm
by topic (cross-sectional)

MMRI trend

MMRI_r time-series per firm
across reporting periods

supports: Investors (disclosure-risk complement to ESG ratings) · Firms (topic-level reporting gaps) · Regulators (scalable alignment monitoring)

The framework converts unstructured narrative ESG disclosures into a structured, comparable, and auditable materiality-misalignment signal.

solid arrows = data flow curved arrow = process repeated per topic ✓ robustness checks = inline evidence of reliability